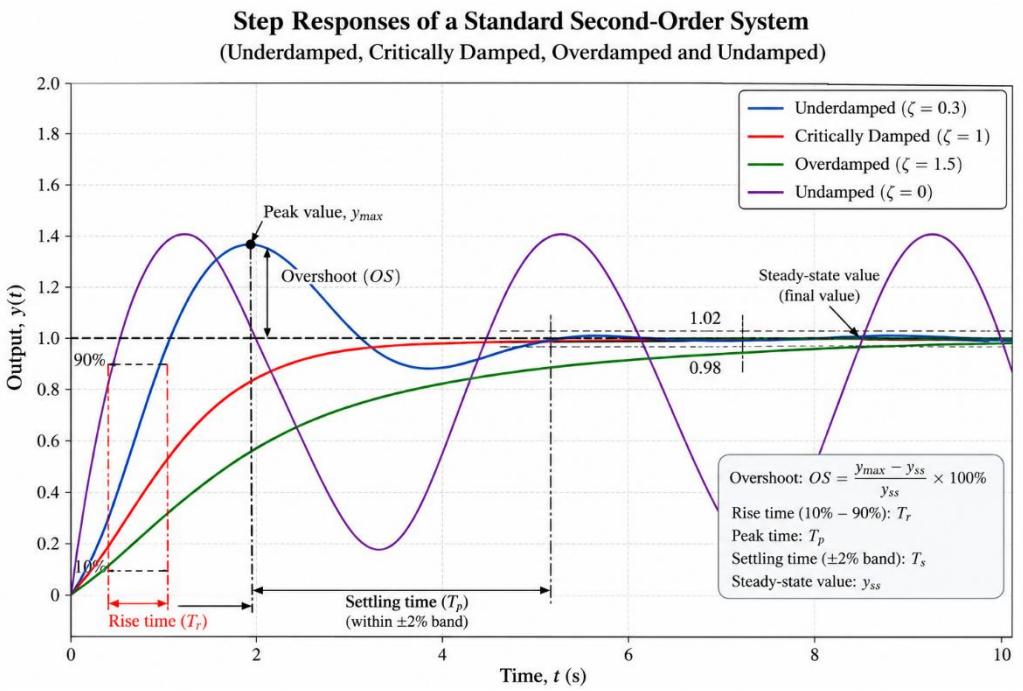
LAB NO: 5

Time Response of Second Order Systems



Objectives

- Derive the transfer function of a series RLC network with capacitor voltage as output.
- Relate R, L, and C to natural frequency, damping ratio, and poles.
- Use MATLAB to calculate poles, overshoot, rise time, and settling time.
- Compare underdamped, critically damped, and overdamped responses.
- Verify analytical predictions and defend modeling decisions in an individual viva.

1. Electrical network model: series RLC circuit

Consider a series RLC circuit driven by input voltage vi(t), with output capacitor voltage vC(t). Assume ideal components and zero initial energy unless otherwise specified.

KVL for the series RLC circuit: $v_i(t) = L\, di(t)/dt + R\, i(t) + v_C(t)$ Capacitor current-voltage relation: $i(t) = C\, dv_C(t)/dt$. Substituting the capacitor relation into KVL gives the governing differential equation:

$$LC\, d^2v_C(t)/dt^2 + RC\, dv_C(t)/dt + v_C(t) = v_i(t).$$

For zero initial conditions, the transfer function is:

$$G(s) = V_C(s)/V_i(s) = 1/(LCs^2 + RCs + 1)$$

Standard second-order form:

$$G(s) = \omega_n^2/(s^2 + 2\zeta\omega_n\, s + \omega_n^2)$$

Natural frequency: $\omega_n = 1/\sqrt{LC}$

Damping ratio: $\zeta = (R/2)\sqrt{(C/L)}$ *The DC gain is unity*: $K_DC = 1$.

Damping classification:

$$\zeta < 1: \textit{underdamped};\quad \zeta = 1: \textit{critically damped};\quad \zeta > 1: \textit{overdamped}$$

The poles are the roots of:

$$s^2 + 2\zeta\omega_n\, s + \omega_n^2 = 0$$

2. MATLAB function

Command	Purpose	Example
tf	Create transfer function	G = tf(1,[L*C R*C 1]);
step	Unit-step response	step(G); grid on;
impulse	Impulse response	impulse(G);
pole	Calculate poles	pole(G)
damp	Natural frequencies and damping	damp(G)
dcgain	Steady-state gain	dcgain(G)
stepinfo	Time-response metrics	stepinfo(G)
pzmap	Pole-zero map	pzmap(G); grid on;
lsim	Custom-input response	lsim(G,u,t);
feedback	Unity negative feedback	T = feedback(G,1);
hold on/off	Overlay plots	hold on; step(G2); hold off;

3. experiment Problems: predict, simulate, verify

Experiment A — Build and inspect an RLC model

Use the following component values: L = 0.5 H, C = 0.02 F, and R = 2 Ω. Before using MATLAB, predict the natural frequency, damping ratio, poles, final value, and overshoot.

```
clc; clear; close all;
R = 2; L = 0.5; C = 0.02;
G = tf(1,[L*C R*C 1]);
wn = 1/sqrt(L*C);
zeta = (R/2)*sqrt(C/L);
G
wn
zeta
pole(G)
damp(G)
dcgain(G)
stepinfo(G)
figure; step(G); grid on;
title('Series RLC: Capacitor Voltage Step Response');
xlabel('Time (s)'); ylabel('v_C(t) / V_step');
```

Expected checks: ωn ≈ 10 rad/s; ζ ≈ 0.2, so the response is underdamped and overshoots. For a unit-step input, final capacitor voltage is 1.

Experiment B — Compare damping ratios

Keep L = 0.5 H and C = 0.02 F. Calculate critical resistance:$R_critical = 2\sqrt{(L/C)}$Then compare an underdamped, critically damped, and overdamped circuit.

```
clc; clear; close all;
L = 0.5; C = 0.02;
Rcrit = 2*sqrt(L/C);
Rvals = [2 Rcrit 2.5*Rcrit];
figure; hold on;
for k = 1:length(Rvals)
    R = Rvals(k);
    G = tf(1,[L*C R*C 1]);
    step(G);
end
grid on;
legend('Underdamped','Critical','Overdamped','Location','southeast');
title('Effect of Resistance on RLC Step Response');
xlabel('Time (s)'); ylabel('Normalized capacitor voltage');
hold off;
```

Increasing R increases damping. The underdamped response oscillates and overshoots. Critical damping is the fastest non-oscillatory case; overdamping is non-oscillatory but slower.

Lab Exercise 1

Construct a transfer function

For R = 4 Ω, L = 1 H, and C = 0.1 F, construct the transfer function:$G(s) = V_C(s)/V_i(s) = 1/(LCs^2 + RCs + 1)$

Tasks: Use tf; display G, poles, DC gain, and plot the step response.

Lab Exercise 2

Identify damping class

For R = 8 Ω, L = 0.5 H, and C = 0.02 F, calculate:$\omega_n = 1/\sqrt{(LC)}, \quad \zeta = (R/2)\sqrt{(C/L)}$

Tasks: Verify using damp(G), then classify the system.

Lab Exercise 3— Design an RLC network

Analyze a given RLC circuit

A series RLC circuit has L = 0.8 H, C = 0.0125 F, and R = 4 Ω. Input is a unit-step voltage; output is capacitor voltage.$G(s) = V_C(s)/V_i(s) = 1/(LCs^2 + RCs + 1)$

1. Derive the transfer function and calculate natural frequency and damping ratio.
2. Use MATLAB to find poles, damping class, DC gain, and stepinfo metrics.
3. Plot the step response and pole-zero map.
4. Verify final value and explain whether overshoot occurs.
5. Explain how R, L, and C appear in the denominator coefficients.

Lab Exercise 4 — Design to transient specifications

Design a series RLC circuit with C = 0.01 F. Required unit-step overshoot is less than 10%, and 2% settling time is no more than 1.5 s. $Tp < 0.10; \quad T_s \leq 1.5\,s$. Use the standard approximation T_s ≈ 4/(ζω_n) for the 2% criterion.$T_s \approx 4/(\zeta\omega_n)$

6. Choose a feasible damping ratio and natural frequency using second-order design relationships.
7. Calculate R and L.

8. Verify overshoot and settling time in MATLAB.
9. If the design fails either requirement, revise the component values and document the change.
10. Explain why satisfying the overshoot constraint does not guarantee the settling-time constraint.